

SMART AMBULANCE TRAFFIC SIGNAL SYSTEM

Aim:

To design and deploy a low-cost, V2X-capable emergency clearance system that automatically grants green signal priority to approaching ambulances, alerts surrounding drivers to clear blind spots, and uses an integrated computer-vision setup to enforce automatic traffic compliance (challans) for non-yielding vehicles.

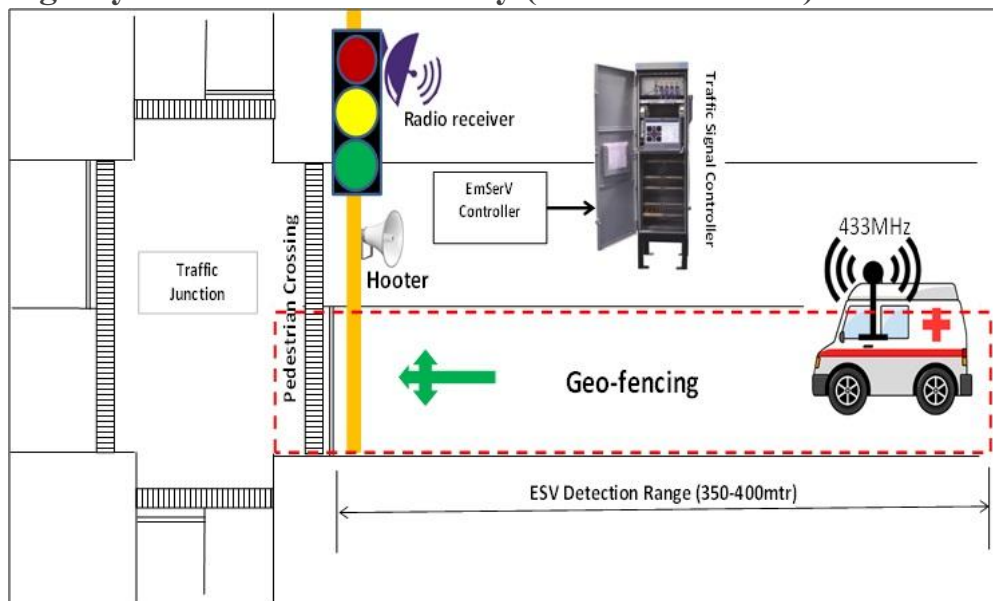
Problem statement:

Rapid urbanization and rising vehicle density have led to severe traffic congestion, particularly during peak commuting hours. While general traffic delays cause everyday inconvenience, they present a life-threatening hazard for emergency vehicles. Ambulances race against time during the critical "Golden Hour," where even a few minutes' delay can decide a patient's survival. Conventional traffic signals operate on fixed time cycles and cannot detect or prioritize approaching ambulances.

Furthermore, drivers ahead often fail to yield due to blind spots, high ambient noise, or poor road discipline. Without automated signal pre-emption, real-time driver alerting, and dynamic enforcement, emergency vehicles remain trapped in congestion, directly jeopardizing patient care.

Evidence

1. Emergency Service Vehicle Priority (C-DAC EmSerV)



“The vehicles cannot negotiate through the traffic when the traffic is stopped in Red. The EmSerV will open all green signals in the direction of the ESV sufficiently before the emergency vehicle reach the junction.”

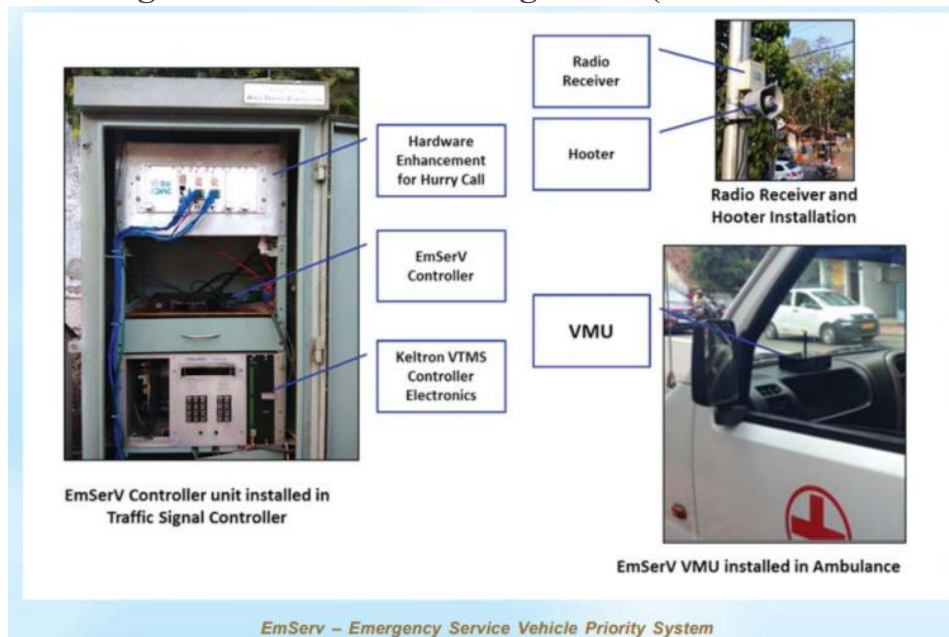
Source: C-DAC, Government of India — EmSerV (<https://www.cdac.in/>)

2. Vehicle Identification & Heading Data Transmission (NASA)

“Street intersections that are equipped with traffic lights would also be equipped with means for generating audible announcements of approaching emergency vehicles, according to a proposal.”

Source: NASA Technical Reports Server — Vehicle Transponder for Pre-emption of Traffic Lights ([NTRS Citation 20110013061](#))

3. Roadside Signal Control & Listening Mode (C-DAC Annual Report)



“The traffic signal controller opens right-of-way for the ESV by terminating the currently running signal phase. The signal sequence is resumed once the ESV clears the junction”

Source: C-DAC Annual Report 2017–18 ([C-DAC Publications](#))

4. Automated Violation & Enforcement Systems (C-DAC iRIDS)

“C-DAC has developed Red Light Violation Detection System (iRIDS) that provides solution to automatic identification of red-light violations at signalized traffic junctions. It automatically captures and presents evidence of the red-light violation that helps smooth functioning of law enforcement.”

Source: C-DAC, Government of India — iRIDS ([Professional Electronics Division](#))

5. V2X Emergency Vehicle Pre-emption & Smart City Integration (Focal Point Positioning)

“In the context of emergency vehicles, V2X communication can adjust traffic systems and provide real-time alerts to both drivers and infrastructure to ensure that first responders can reach their destinations quickly and safely.”

Source: Focal Point Positioning — V2X for Safety: Vehicle-to-Everything Technology for Emergency Vehicle Management in Smart Cities ([Technical Article](#))

6. National Standards for V2V & Emergency Vehicle Alerts (MoRTH / PIB India)

“V2V communication enables nearby vehicles to exchange real-time information, including their speed, position, direction, acceleration, etc. This information can provide advance warnings to drivers or vehicle systems in safety-critical situations, including sudden braking, forward-collision risk, unsafe lane changes and the approach of emergency vehicles.”

Source: Press Information Bureau (PIB), Government of India — Vehicle-to-Vehicle Communication Systems (Release ID: 2293692) ([PIB Press Release](#))

7. Integrated RFID/Wireless Traffic Override Research (UNIMAS TUR)

“The traffic lights can be controlled in a timely and efficient manner every time an emergency ambulance is approaching. “

Source: *Trends in Undergraduate Research — Smart Ambulance Traffic Control System (UNIMAS)*
([TUR Journal Article](#))

8. Poor lane discipline

People lacked awareness to give way to an ambulance during an emergency. People were lacking how to behave on the roads; they did not follow the lane system.



9. High death rates due to traffic congestion

Generally, ambulances in India are fast, but due to low cooperation between people to give way to ambulance and getting stuck at traffic lights cause many deaths per day. According to the National Crime Records Bureau, nearly 24,012 people die each day due to a delay in getting medical assistance.



Sources: <https://www.thehindu.com/news/cities/mumbai/helping-patients-get-to-hospital-on-time/article18031300.ece>

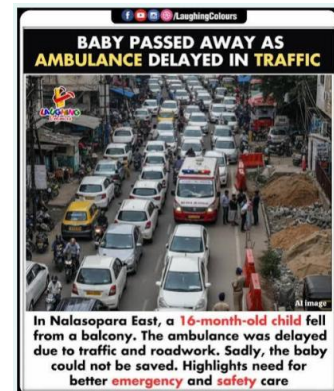
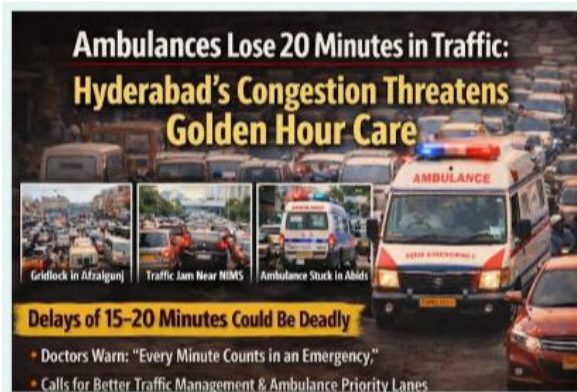
<https://radhee.com/advocacy/morth-road-safety/>

10. Other evidences

Sources:

<https://hellohyderabad.org/ambulances-lose-20-minutes-in-traffic-hyderabad-congestion-threatens-golden-hour-care/>

<https://www.indiatoday.in/india/kerala/story/traffic-jam-in-cozhikode-kerala-claims-lives-of-two-patients-in-ambulances-2657470-2024-12-30>



Conclusion

The **Smart Ambulance Traffic Signal System** merges affordable hardware prototyping (433 MHz RF, magnetometers, and Arduino controllers) with verified national and international technological standards. Literature and policy documentation from **C-DAC**, **NASA**, **MoRTH (Government of India)**, and **IEEE/ITS research** proves that V2I traffic pre-emption, V2V driver alerting, automated fine enforcement, and dynamic signal override are established solutions for modern urban emergency transit.